

FPX Online Banking Gateway Assessment Brief

Organization: Synthetic PayNet E-Commerce Rail Lab

Scenario: FPX Online Banking Gateway

Classification: synthetic test evidence only.

Public inspiration: FPX is publicly described as a real-time online payment gateway for e-commerce and bank-account payments.

System summary: Synthetic merchant checkout, bank selection, bank redirect, authorisation, debit confirmation, and merchant receipt notification flow.

Risk goal: Critical merchant checkout and bank redirect exposure with classical crypto in high-value flows.

Key findings: TLS 1.2 RSA key exchange at merchant redirect gateway; ECDH bank session bridge; RSA-3072 signed payment confirmation payloads; TLS 1.3 ML-KEM pilot notification endpoint.

Expected scan behavior: Public merchant checkout endpoints and bank redirect endpoints should elevate TLS 1.2 RSA/ECDH findings; internal AES-256 and SHA-384 controls should remain low risk.

Recommended validation: disable TLS 1.2 RSA suites, prepare participant bridge certificates for ML-DSA, and retain AES-256 ledger controls.

No credentials, private keys, payment tokens, PINs, account numbers, customer data, or live infrastructure references are included.